SDC HACKATHON · TEAM ARRAY FIVE

SCDC Recovery

When a supplier lets a small business down, see what breaks, compare fixes, and decide in minutes.

AI proposes · the rules verify · a human decides

SDG 8: Decent Work and Economic Growth

WHO IT'S FOR

Meet Seri Kenanga Events

A small event planner in Kuala Lumpur. Weddings and seminars, a handful of suppliers, and everything runs on WhatsApp and one spreadsheet.

Nurul

Owner. Makes every call.

Hafiz

Staff. Logs problems as they come in.

Demo business. Fictional, but typical.

AND THEY ARE THE NORM

96.9%

of business establishments in Malaysia are MSMEs, about 1.1 million firms.

Source: SME Corp Malaysia, Economic Census 2023

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THE PROBLEM

One late delivery. Every job that depends on it is now at risk, and nobody has the full picture.

Which jobs are hit?

Owners check every booking by hand, across chats and spreadsheets.

Which deadlines slip?

Shared stock means one delay can quietly break two events.

What are the options?

Other suppliers, spare stock, lead times: all in someone’s head.

Who do I tell?

Customers and suppliers need clear messages, fast.

OUR SOLUTION

From a supplier’s WhatsApp to an approved plan, in five steps

1

Report

Maju messages on WhatsApp; the report files itself (or paste a message into the form).

2

See impact

Every affected job, and why.

3

Compare

Recovery options, each checked by rules.

4

Approve

Only the owner can decide.

5

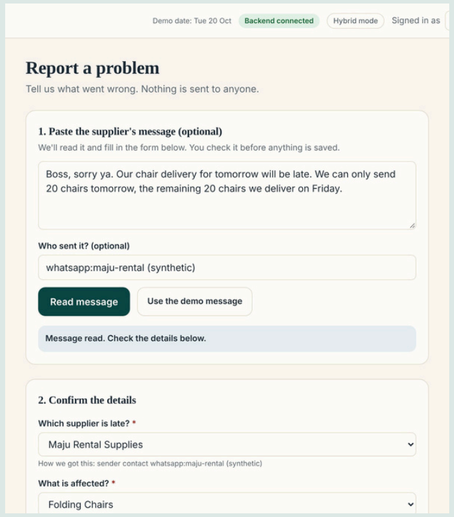
Follow up

Draft messages and a full log.

LIVE DEMO

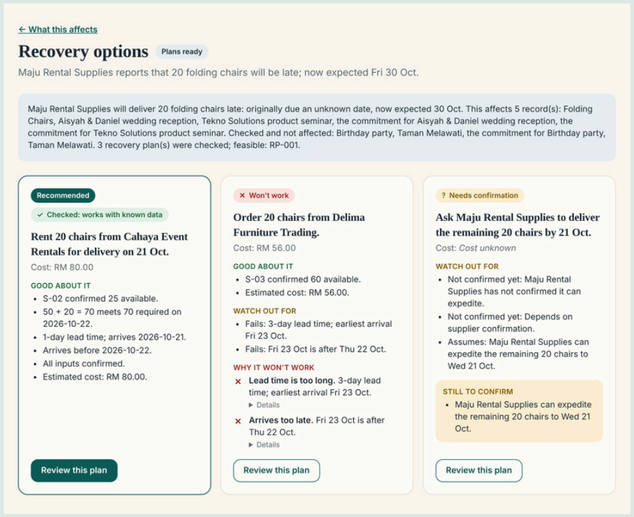
Report. Compare. Approve.

1 Report



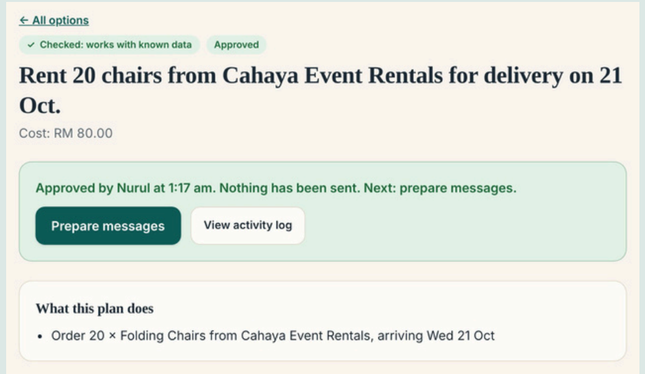
Maju messages on WhatsApp: only 20 of 40 chairs arrive. The report files itself (or paste a message into the form).

2 Compare



Short 20 chairs on Thursday. Three fixes, each checked: Cahaya for RM 80 works, Delima is too late.

3 Approve



Only Nurul can approve. The app drafts the supplier message and logs every step.

HOW IT WORKS

AI reads and explains. Rules decide what’s true.



AI never decides feasibility and never writes to the database. Every plan is re-checked by deterministic code, even after the owner edits it. If the AI service is down, templates take over and the app keeps working.

All demo data is synthetic. In production: role-based access, minimum data (PDPA), nothing sent automatically.

WHAT WE BUILT

A working prototype, tested end to end

656

automated tests passing

Backend, frontend and end-to-end, 24 October. 583 backend + 73 frontend.

Frontend	React, TypeScript, Vite, Tailwind. Mock, hybrid and real API modes.
Backend	Python, FastAPI and SQLAlchemy. Roles, approvals, activity log.
Impact engine	Dependency graph, quantity and deadline checks, evidence for every result.
Recovery planner	Candidate plans, rule-based validation and ranking.
AI module	Message reading, explanations, drafts. Swappable, with a template fallback.

WHY IT MATTERS

Faster, safer decisions for owners who can’t afford a miss

- No affected job slips through the cracks
- Options are checked before anyone commits money
- The owner stays in control, with a record of every decision

What’s next

- Pilot with 5 event planners in the Klang Valley
- Connect WhatsApp so messages arrive on their own
- More industries: catering, repairs, logistics
- Staff absence and broken equipment, not only late suppliers

BUILT FOR THE ORBYTE HUB

Message reader, impact engine and recovery planner are reusable modules any supplier-dependent SME can plug into.

Freemium, then a monthly fee per business via Orbyte.

THE TEAM

ARRAY-FIVE: five people, five workstreams

TEAM LEADER

Yosef Kristian Tirta

Frontend and user experience

Nicholas Dillon

Backend, database and API

Joshua Michael N. Tjoa

Dependency graph and impact analysis

Raymond Russel Willim

Recovery planning and validation

Erwin Okto Wu

AI integration and evaluation

AI proposes. The rules verify. A human decides.

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https://array-five.replit.app · Thank you